



ANNUAL REPORT

2025

Joint Center for Nuclear Physics
青年核物理中心

Contents

1. Next-to-leading-order prediction for the neutrinoless double-beta decay	
Y. L. Yang and P. W. Zhao	
Physical Review Letters 134 (2025) 242502	
.....	1
2. Chiral symmetry and peripheral neutron-α scattering	
Y. L. Yang, E. Epelbaum, J. Meng, L. Meng, and P. W. Zhao	
Physical Review Letters 135 (2025) 172502	
.....	7
3. Machine learning orbital-free density functional theory resolves shell effects in deformed nuclei	
X. H. Wu, Z. X. Ren, and P. W. Zhao	
Communications Physics 8 (2025) 316	
.....	14
4. Correlation between U/Th and Pb/Os abundance ratios and its application in nuclear cosmochronology	
Y. Y. Huang, Q. Q. Cui, X. H. Wu, and S. Q. Zhang	
The Astrophysical Journal 988 (2025) 22	
.....	23
5. Exact-exchange relativistic density functional theory in three-dimensional coordinate space	
Q. Zhao, Z. X. Ren, P. W. Zhao, and K. Yoshida	
Physics Letters B 860 (2025) 139196	
.....	30
6. Relativistic low-momentum interactions from renormalization group	
T. X. Huang, Y. L. Yang, J. Meng, P. Ring, and P. W. Zhao	
Physics Letters B 866 (2025) 139502	
.....	36
7. Intrinsic spin distributions in multinucleon transfer reactions	
D. D. Zhang, D. Vretenar, T. Nikšić, P. W. Zhao, and J. Meng	

.....	42
8. Shape evolution in neutron-rich Rh isotopes: First measurement of negative-parity isomers in $^{117,119}\text{Rh}$	
J. Z. Zhang, Z. Q. Chen, T. Qu, Y. K. Wang, Z. H. Li, H. Hua, P. W. Zhao, H. Watanabe, G. Lorusso, C. X. Yuan, S. Nishimura, H. Baba, G. Benzoni, F. Browne, K. Y. Chae, F. C. L. Crespi, P. Doornenbal, N. Fukuda, R. Gernhäuser, G. Gey, C. Y. Guo, N. Inabe, T. Isobe, D. X. Jiang, Y. Jin, H. S. Jung, A. Jungclaus, D. Kameda, G. D. Kim, Y. K. Kim, I. Kojouharov, F. G. Kondev, T. Kubo, N. Kurz, G. J. Lane, X. Q. Li, J. L. Lou, D. W. Luo, A. Montaner-Pizá, K. Moschner, F. Naqvi, L. Ni, M. Niikura, H. Nishibata, A. Odahara, R. Orlandi, Z. Patel, Zs. Podolyák, H. Sakurai, H. Schaffner, G. S. Simpson, P.-A. Söderström, K. Steiger, T. Sumikama, H. Suzuki, H. Takeda, J. Taprogge, A. Wendt, H. Y. Wu, J. Wu, C. Xu, Z. Y. Xu, A. yagi, Y. L. Ye, K. Yoshinaga, S. Q. Zhang, S. Y. Zhang, Z. X. Zhou	
Physics Letters B 873 (2026) 140144	
.....	48
9. Neutrinoless double-beta decay in a finite volume from relativistic effective field theory	
Y. L. Yang and P. W. Zhao	
Physical Review D 111 (2025) 014507	
.....	57
10. Reconciling light nuclei and nuclear matter: relativistic <i>ab initio</i> calculations	
Y. L. Yang and P. W. Zhao	
Chinese Physics Letters 42 (2025) 051201	
.....	74
11. Nuclear matter in relativistic Brueckner–Hartree–Fock theory using the leading order covariant chiral interactions with local and nonlocal regulators	
W. J. Zou, Y. L. Yang, S. H. Shen, and J. Meng	
Chinese Physics Letters 42 (2025) 080101	
.....	80
12. Time-dependent relativistic Hartree-Fock model with spherical symmetry	
J. Geng, Z. H. Wang, P. W. Zhao, Y. F. Niu, H. Z. Liang, and W. H. Long	
Physical Review C 111 (2025) 024305	
.....	85

13. Speed of sound bounds and first-order phase transitions in compact stars

P. L. Patkos, G. A. Lalazissis, S. B. Wang, J. Meng, P. Ring, and Ch. C. Moustakidis

Physical Review C **111** (2025) 025801

.....95

14. Ground-state properties and structure evolutions of odd- A transuranium Bk isotopes from deformed relativistic Hartree-Bogoliubov theory in continuum

Z. D. Huang, W. Zhang, S. Q. Zhang, and T. T. Sun

Physical Review C **111** (2025) 034314

.....106

15. Microscopic model for yields and total kinetic energy in nuclear fission

B. Li, D. Vretenar, T. Nikšić, P. W. Zhao, and J. Meng

Physical Review C **111** (2025) L051302

.....117

16. Relativistic configuration-interaction density functional theory for nuclear wobbling motion

T. Qu, Y. K. Wang, and P. W. Zhao

Physical Review C **111** (2025) 064309

.....123

17. Charge radii and their deformation correlation for even- Z nuclei in deformed relativistic Hartree-Bogoliubov theory in continuum

C. Pan and J. Meng

Physical Review C **112** (2025) 024316

.....131

18. Charge radii of calcium isotopes within relativistic density functional theory: The finite size of the nucleon and quadrupole shape-fluctuation effects

H. H. Xie, J. Li, Y. L. Yang, and P. W. Zhao

Physical Review C **112** (2025) L021303

.....140

19. First observation of pseudospin-chiral triplet bands with pure proton configuration

W. B. Zeng, C. Liu, S. Y. Wang, B. Qi, R. A. Bark, H. Jia, J. Meng, S. Q. Zhang, P. Jones, S. M. Wyngaardt, Q. B. Chen, Z. Q. Li, S. Wang, D. P. Sun, R. J. Guo, X. C. Han, W. Z. Xu, H. Hua, X. Q. Li, C. G. Li, R. Han, B. H. Sun, L. H. Zhu, T. D. Bucher, J. Ndayishimye, J. F. Sharpey-Schafer, O. Shirinda, M. Wiedeking, T. Dinoko, S. S. Ntshangase

Physical Review C **112** (2025) 034330

.....147

20. Examining the characteristics of chiral-parity violation in nuclear systems: Influence of triaxial deformation

Y. Y. Wang and S. Q. Zhang

Physical Review C **112** (2025) 034334

.....155

21. Octupole correlations, pseudospin partner bands, and γ vibrational band in ^{75}As

C. Liu, W. B. Zeng, S. Y. Wang, H. Zhang, B. Qi, X. Xiao, R. A. Bark, J. Meng, S. Q. Zhang, H. Hua, P. Jones, S. M. Wyngaardt, W. Z. Xu, H. F. Bai, S. Wang, D. P. Sun, Z. Q. Li, H. Jia, R. J. Guo, X. C. Han, L. Mu, X. Lu, C. Y. Niu, C. G. Wang, E. A. Lawrie, J. J. Lawrie, J. F. Sharpey-Schafer, M. Wiedeking, S. N. T. Majola, T. D. Bucher, T. Dinoko, B. Maqabuka, L. Makhathini, L. Mdletshe, N. A. Khumalo, O. Shirinda, K. Sowazi

Physical Review C **112** (2025) 034337

.....165

22. Triaxial relativistic Hartree-Bogoliubov theory in continuum for exotic nuclei

K. Y. Zhang, S. Q. Zhang, and J. Meng

Physical Review C **112** (2025) 044308

.....174

23. Negative-parity high-spin structure of ^{105}Pd

B. Kruzsicz, D. Sohler, J. Timár, I. Kuti, Q. B. Chen, S. Q. Zhang, J. Meng, P. Joshi, R. Wadsworth, K. Starosta, A. Algora, P. Bednarczyk, D. Curien, Zs. Dombrádi, G. Duchéne, A. Gizon, J. Gizon, D. G. Jenkins, T. Koike, A. Krakó, A. Krasznahorkay, J. Molnár, B. M. Nyakó, E. S. Paul, G. Rainovski, J. N. Scheurer, A. J. Simons, C. Vaman, L. Zolnai

Physical Review C **112** (2025) 064316

.....197

24. Exploratory study on the masses of odd- Z nuclei and r -process simulation based on the deformed relativistic Hartree-Bogoliubov theory in continuum

C. Pan, Y. C. Yang, X. F. Jiang, and X. H. Wu

The European Physical Journal A 61 (2025) 275	210
25. Octupole correlations in stable nucleus ^{153}Eu within reflection-asymmetric particle rotor model	
Y. Y. Wang and S. Q. Zhang	
Chinese Physics C 49 (2025) 014103	226
26. Tilted-axis-cranking covariant density functional theory for high-spin spectroscopy of ^{69}Ga	
Y. P. Wang, Y. K. Wang, and P. W. Zhao	
Chinese Physics C 50 (2026) 014112	233
27. Relativistic <i>ab initio</i> calculations for static and rotating neutron stars	
H. Tong, S. B. Wang, and J. Meng	
Frontiers in Astronomy and Space Sciences 12 (2025) 1666331	238
28. Multiple chiral doublet bands in ^{104}Rh	
A. Krakó, D. Souler, J. Timár, I. Kuti, Q. B. Chen, S. Q. Zhang, J. Meng, K. Starosta, T. Koike, E. S. Paul, D. B. Fossan, and C. Vaman	
Acta Physica Polonica B Proceedings Supplement 18 (2025) 2-A40	247
29. Halo nuclei initializing a new era for exotic nuclei study	
W. H. Long, J. Meng, and B. H. Sun	
Chinese Science Bulletin 70 (2025) 3211-3213	254
30. Progress of nuclear theory on studying the neutrinoless double-β decay	
D. L. Fang, J. Meng, Y. F. Niu, Z. M. Niu, Y. K. Wang, J. M. Yao, and P. W. Zhao	
SCIENTIA SINICA Physica, Mechanica & Astronomica 55 (2025) 111007	257
31. Relativistic configuration–interaction density functional theory	
Y. K. Wang and J. Meng	

International Journal of Modern Physics E **34** (2025) 2543004

.....**274**

32. Maximally stiff equations of state and the structure of hybrid stars

P. L. Patkos, G. A. Lalazissis, S. B. Wang, J. Meng, P. Ring, and Ch. C. Moustakidis

HNPS Advances in Nuclear Physics **31** (2025) 8151

.....**283**